	AzureSky	
	Method Validation for Calorific Value Determination	Document no: AS002MV
Compiled by:	Sifiso Nyambi - Laboratory Manager	
Approved by:	Nonhlanhla Msibi -Operations Manager	

Method Validation Report

Measurement Method: ISO 1928

Reference Material (RM): GA CRM AS002

Date of Characterisation: May 2025

Analyst: Keitumetse

Number of Replicates: 10 units × 2 replicates

1. Introduction

This report presents the validation of the method applied for the determination of calorific value in solid fuel samples, in accordance with ISO 1928. The objective of validation is to confirm that the method provides reliable, accurate, and precise results under repeatability, reproducibility, and robust conditions.

Key parameters assessed include:

- Precision (repeatability and reproducibility)
- Accuracy (trueness and recovery)
- Robustness/ruggedness (temperature effect)
- Homogeneity of the reference material
- Measurement uncertainty

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2. Methodology

- Sample mass range: 0.5000–0.5099 g
- Replicates: 8–10 replicate determinations per analyst per condition
- Assigned value (reference): 24.07 MJ/kg (based on benzoic acid calibration AV = 26.43 MJ/kg)
- Acceptance criteria:
  - Difference between measured and assigned value ≤0.30 MJ/kg
  - Recovery: 98–102%
  - Precision (RSD): ≤1%

Data was collected from three analysts under normal and elevated temperature conditions, followed by statistical analysis (t-tests, ANOVA, homogeneity testing, and uncertainty estimation).

3. Results

3.1 Precision (Repeatability and Reproducibility)

Analyst	Average (MJ/kg)	SD	RSD (%)	Absolute Difference from AV	Recovery (%)
1	23.93	0.09	0.39	0.14	99.42
2	23.77	0.05	0.21	0.30	98.76
3	23.93	0.05	0.21	0.14	99.42

RSD values are all <0.5%, confirming excellent repeatability and reproducibility. Absolute differences ≤0.30 MJ/kg, within acceptance. Recoveries between 98.76–99.42%, within acceptance.

Conclusion: Precision and accuracy requirements are met.

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3.2 Ruggedness / Robustness (Temperature Effect)

ANOVA was used to compare results under normal lab conditions (19–26 °C) and elevated conditions (27.56 °C).

- Normal average: 23.93 MJ/kg (variance 0.0088)
- Elevated average: 23.88 MJ/kg (variance 0.0052)
- F-stat = 1.72 < F-critical = 4.60; p = 0.21 > 0.05

  Conclusion: No statistically significant difference between conditions. Method is robust to small environmental temperature changes.

3.3 Homogeneity Assessment

10 units × 2 replicates were analysed. ANOVA showed:

- Fcalc = 0.68 < Fcrit = 3.02
- No significant between-unit variation detected.
- Homogeneity confirmed.

3.4 Measurement Uncertainty

3.4.1 Sources of Uncertainty

Source	Symbol	Type	Std. Uncertainty (U)	Distribution	Divisor	Sensitivity	U^2
Repeatability	U1	A	0.12	Normal	1	1	0.0144
Balance calibration	U2	B	0.0157	Rectangular	√3	1	0.00025
Homogeneity	U3	A	0.1083	Normal	1	1	0.01174
Stability	U4	B	NA	Rectangular	√3	1	-
PT Characterisation	U5	B	NA	Normal	1	1	-

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3.4.2 Combined and Expanded Uncertainty

- Combined Uncertainty (Uc): 0.14 MJ/kg
- Expanded Uncertainty (U, k=2): 0.28 MJ/kg
- Coverage probability: 95%

4. Discussion

- Precision: Both intra-analyst (repeatability) and inter-analyst (reproducibility) results show RSD <0.5%, which is well within acceptance limits.
- Accuracy: Absolute differences ≤0.30 MJ/kg and recoveries close to 100% demonstrate trueness.
- Ruggedness: Method performance is not significantly affected by moderate environmental temperature variations.
- Homogeneity: CRM units show no significant between-unit differences, confirming reliability. Measurement uncertainty: The final expanded uncertainty of 0.28 MJ/kg is consistent with typical performance for ISO 1928 calorific value determination.

5. Conclusion

- The method validated in this study is:
- Precise (RSD <0.5%)
- Accurate (recoveries 98.8–99.4%)
- Robust (insensitive to small environmental changes)
- Homogeneous (CRM units consistent)
- Traceable and reliable (expanded uncertainty 0.28 MJ/kg, 95% CL)

Therefore, the method meets ISO 1928 requirements and is fit for purpose for the determination of calorific value in solid fuel samples.

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6. Final Reporting Statement

The calorific value determination by ISO 1928 method has an expanded uncertainty of 0.28 MJ/kg (k=2, 95% confidence level). All results are traceable to SI units through calibrated reference materials and instruments.

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